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Research Proposal

Subject: Artificial Intelligence / Psychology

Algorithmic Mirror

Does the Disclosure of an AI-Generated Personality Description Affect Self-Concept and Behaviour?

Full Legal Name: Nancy {my full legal name in my documents is said to be Just my Front name}

Institution: Amity University

Category: Senior (University 1st Year to Any Age)

Country (as registered): India

Course / Year: PhD, Year 1

Discipline: Neuropsychology

Date: 21/09/2026

Abstract

Artificial intelligence systems increasingly infer personality characteristics from digital behaviour, such as language use, application activity, and communication patterns, and then communicate these inferences directly to users through wellness apps, dating platforms, career-assessment tools, and conversational agents. This proposal introduces the Algorithmic Mirror Model, which proposes that an AI-generated personality description can be internalised into a person's self-concept, shape subsequent behaviour, and thereby generate new data that appears to confirm the AI's original assessment — a potential feedback loop between machine inference and human identity. A mixed-method, longitudinal experiment is proposed to test whether receiving an AI-generated personality description produces measurable change in self-concept and behaviour, and whether this effect depends on the perceived credibility of the source. The proposal also considers the ethical implications of psychologically influential AI feedback, particularly for adolescents and young adults.

Keywords: artificial intelligence, personality computing, self-concept, self-fulfilling prophecy, algorithmic identity

1. Problem Statement

Personality assessment traditionally required active participation — a questionnaire, an interview, or a clinician's interpretation. Today, personality-related conclusions can instead be inferred continuously and passively from digital behaviour, and are communicated directly to users with an appearance of data-driven objectivity that a horoscope or a one-off quiz does not carry [1], [2]. This raises a question that goes beyond predictive accuracy: what happens psychologically when a system that appears authoritative repeatedly tells a person who they are? Classic findings on expectation effects [3] and on the tendency to accept general personality statements as uniquely accurate [4] suggest that such disclosures are unlikely to be psychologically neutral, particularly when repeated and framed as evidence-based.

2. Literature Review

Personality computing has established that traits can be estimated from passive digital signals; a meta-analysis of smartphone-derived behavioural data reported meaningful predictive accuracy for Big Five traits [1], and more recent work shows that large language models can infer psychological dispositions from unstructured text with correlations to validated self-report measures, albeit with variable accuracy [2]. These studies address whether AI can predict personality, not what happens once a prediction is disclosed to the person it describes.

Social psychology offers relevant mechanisms. The Barnum effect describes the tendency to rate generic, favourable personality statements as personally accurate [4], a pattern also documented in relation to ego identity and mental health among adolescents receiving personality feedback [5]. The self-fulfilling prophecy, from Merton's original formulation [6] through the Pygmalion effect in expectation research [3], shows that an expectation communicated to a person can alter behaviour in ways that appear to validate the expectation. Cooley's looking-glass self [7] similarly holds that self-concept develops partly through perceived external evaluation — a role an AI system may increasingly occupy.

A separate literature on algorithmic identity argues that repeated interaction with algorithmic systems can actively shape self-understanding rather than merely reflect it [8]; qualitative work on recommendation feeds shows users recognising identity in algorithmically curated content and adjusting behaviour in response [9], and research on filter bubbles describes a related loop in which

online expression and algorithmic amplification reinforce one another [10]. Personality computing, expectation-based social psychology, and algorithmic-identity research have developed largely in isolation, leaving open whether an AI-generated personality prediction can itself change the self-concept and behaviour it claims only to describe.

3. Research Question and Theoretical Model

Primary question: Does receiving an AI-generated personality description produce measurable change in self-concept and subsequent behaviour, relative to no disclosure? Secondary question: Does this effect vary with the perceived credibility of the disclosing source?

The Algorithmic Mirror Model proposes four stages: (1) Inference — an AI system analyses digital behaviour and generates a personality profile; (2) Disclosure — the profile is communicated directly and personally to the individual; (3) Internalisation — the individual incorporates the description into self-concept, especially when it is positive, ambiguous, or attributed to a credible source, and begins behaving in line with it; (4) Re-inference — the system observes the resulting behaviour as new data consistent with its original prediction, closing a potential feedback loop between machine inference and lived identity.

Four predictions follow: (a) AI-disclosed descriptions will shift scores on validated personality measures toward the disclosed profile; (b) the effect will be stronger when the source is believed to be AI-derived rather than human-derived; (c) positive, ambiguous descriptions will shift self-concept more than specific negative ones; (d) repeated exposure will produce larger cumulative change than a single exposure.

4. Proposed Methodology

A mixed-method, longitudinal experimental design is proposed. Participants first complete a validated Big Five measure, then are randomly assigned to one of three conditions: (1) AI Disclosure — told the profile was generated by AI from their behavioural data; (2) Human Disclosure — given an identical profile attributed to a human psychologist, isolating the effect of perceived source; (3) No-Disclosure Control — no profile is provided. The personality measure is re-administered at one week and one month, alongside behavioural proxy measures where feasible, and participants rate the perceived accuracy and authority of the profile they received.

Participants and sampling. Adults aged 18 or over will be recruited through university and online channels using purposive sampling. A target of about 150 participants (approximately 50 per condition) is proposed, with the final sample size confirmed by an a priori power analysis for the smallest effect of interest.

Measures and timeline. The baseline and follow-up personality measure will be the Big Five Inventory-2 (BFI-2) [11]. To allow predictions (c) and (d) to be tested, profile favourability (positive-ambiguous or specific-negative, but mild and non-harmful) will be varied across participants within the disclosure conditions, and the profile will be re-presented at the one-week session. Behavioural proxies will be pre-specified measures linked to the disclosed trait, such as choices in a brief follow-up task.

A subgroup will complete semi-structured interviews exploring how participants interpreted the description, whether it felt self-descriptive, whether they noticed or deliberately enacted behavioural change, and whether they attributed any change to the AI feedback. Quantitative change will be analysed with mixed-effects models incorporating condition, favourability of the label, perceived source authority, baseline personality, and age. Interview transcripts will be analysed using thematic analysis [12], and the quantitative and qualitative findings will be integrated through triangulation to

examine whether participants' reported experiences help explain observed change. Any use of deception in the disclosure procedure will be followed by full debriefing, and the study will proceed only after institutional ethical approval.

Ethical safeguards. Participants will give informed consent, may withdraw at any time without penalty, and their data will be anonymised and kept confidential. Profiles will be mild and non-harmful, and the debriefing will explain the true source and purpose of the profile.

5. Significance, Innovation, and Ethical Considerations

AI-based personality tools now operate in domains — mental wellness, dating, education, career guidance, recruitment — where a mistaken or overly confident label could shape real decisions and self-understanding, with adolescents and young adults, whose identities are still forming, particularly exposed [5]. This proposal's contribution is to connect personality computing, expectation-based social psychology, and algorithmic-identity research into a single testable feedback model, extending recent evidence that personality can change through deliberate intervention [13] to the largely unexamined case of AI-driven, ostensibly descriptive feedback. Current consent practices for such tools emphasise data privacy but rarely address the possibility that the feedback itself is psychologically influential; this study supports the case for communicating predictive uncertainty, avoiding overconfident language, and adding safeguards for younger users, without arguing against AI-based personality tools as such.

6. Expected Outcomes, Limitations, and Practicalities

Expected outcomes. We anticipate that AI-attributed profiles will shift Big Five scores toward the disclosed profile more than human-attributed or no-disclosure conditions, with larger shifts for favourable, ambiguous profiles and after repeated exposure.

Limitations. As a pilot-scale experiment, the study relies on self-report measures that may reflect demand characteristics, and a one-month follow-up cannot show lasting identity change. Findings from an adult, largely university-based sample cannot be generalised to adolescents, for whom additional safeguards would be needed before any future work.

Practicalities and post-study plan. The study can be run online with standard survey tools and a small research team after ethical approval. Findings will be shared as a brief on communicating predictive uncertainty in AI personality tools, and will inform a follow-up study on repeated, real-world AI feedback.

7. Conclusion

As AI systems move from analysing behaviour to directly telling people who they are, a new question emerges: does such a description merely reflect an existing personality, or can it also help produce the personality it claims to describe? The Algorithmic Mirror Model frames this as a testable feedback loop — inference, disclosure, internalisation, and re-inference — and the proposed study offers an empirical route to examining it. The result would shift the guiding question for applied AI personality systems from “How accurate is the prediction?” to “What happens when people believe it?” — a question with growing relevance across mental health, education, relationships, and everyday self-understanding.

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